MSG No. 111.257 $P_C\bar{4}2m$ [Type IV, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-42m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `-42m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm`

No.	position	mapping
1	$[0, 0, z]$	`[1,4,7,8]`
2	$[0, 0, -z]$	`[2,3,5,6]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[9,12,15,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[10,11,13,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,4,15,16]`
2	$\left[0, \frac{1}{2}, -z\right]$	`[2,3,13,14]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[5,6,10,11]`
4	$\left[\frac{1}{2}, 0, z\right]$	`[7,8,9,12]`

Table 7: Wyckoff site: `4g`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,7,12,16]`
2	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[2,6,11,13]`
3	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[3,5,10,14]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	`[4,8,9,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `.2.`

No.	position	mapping
1	$[0, y, 0]$	`[1,3]`
2	$[0, -y, 0]$	`[2,4]`
3	$[y, 0, 0]$	`[5,8]`
4	$[-y, 0, 0]$	`[6,7]`
5	$\left[\frac{1}{2}, y + \frac{1}{2}, 0\right]$	`[9,11]`
6	$\left[\frac{1}{2}, \frac{1}{2} - y, 0\right]$	`[10,12]`
7	$\left[y + \frac{1}{2}, \frac{1}{2}, 0\right]$	`[13,16]`
8	$\left[\frac{1}{2} - y, \frac{1}{2}, 0\right]$	`[14,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0, y, \frac{1}{2}\right]$	`[1,3]`
2	$\left[0, -y, \frac{1}{2}\right]$	`[2,4]`
3	$\left[y, 0, \frac{1}{2}\right]$	`[5,8]`
4	$\left[-y, 0, \frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{2}, y + \frac{1}{2}, \frac{1}{2}\right]$	`[9,11]`
6	$\left[\frac{1}{2}, \frac{1}{2} - y, \frac{1}{2}\right]$	`[10,12]`
7	$\left[y + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,16]`
8	$\left[\frac{1}{2} - y, \frac{1}{2}, \frac{1}{2}\right]$	`[14,15]`

Table 10: Wyckoff site: `8j`, site symmetry: `..m`

No.	position	mapping
1	$[x, x, z]$	`[1,8]`
2	$[x, -x, -z]$	`[2,5]`
3	$[-x, x, -z]$	`[3,6]`
4	$[-x, -x, z]$	`[4,7]`
5	$\left[x + \frac{1}{2}, x + \frac{1}{2}, z\right]$	`[9,16]`
6	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -z\right]$	`[10,13]`
7	$\left[\frac{1}{2} - x, x + \frac{1}{2}, -z\right]$	`[11,14]`
8	$\left[\frac{1}{2} - x, \frac{1}{2} - x, z\right]$	`[12,15]`

Table 11: Wyckoff site: 8k, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	[1,16]
2	$\left[x,\, \frac{1}{2}-x,\, -z\right]$	[2,13]
3	$\left[-x,\, x+\frac{1}{2},\, -z\right]$	[3,14]
4	$\left[-x,\, \frac{1}{2}-x,\, z\right]$	[4,15]
5	$\left[x+\frac{1}{2},\, -x,\, -z\right]$	[5,10]
6	$\left[\frac{1}{2}-x,\, x,\, -z\right]$	[6,11]
7	$\left[\frac{1}{2}-x,\, -x,\, z\right]$	[7,12]
8	$\left[x+\frac{1}{2},\, x,\, z\right]$	[8,9]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[x,\, -y,\, -z]$	[2]
3	$[-x,\, y,\, -z]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[y,\, -x,\, -z]$	[5]
6	$[-y,\, x,\, -z]$	[6]
7	$[-y,\, -x,\, z]$	[7]
8	$[y,\, x,\, z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[9]
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[12]
13	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, -z\right]$	[13]
14	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, -z\right]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z\right]$	[16]